Building

The Buildings are the structures that are built with BuildingElements. It is used for collecting and counting the BuildingElements and it is used for some algorithms' calculation.

Because of that every building algorithms' has different needs every building algorithms has its own unique type of Buildings:

- Advanced Grid Building
- Basic Grid Building
- Free Building
- SnapPoint Based Building (Experimental!)

All of them are inherited from the base Building class which is inherited from the MonoBehavior.

The Object structure in your hierarchy should follow the following rules to ensure that the Advanced Building System can work properly.

1. Every BuildingElement is assigned with a building algorithm and every Building should have only BuildingElements with the matching type of algorithm.
2. Every Building should be under a GameObject with BuildingParent script in the Hierarchy and every object under a Building should be a GameObject with Building script.
3. Every BuildingElement should be under a GameObject with Building script in the Hierarchy and every object under a Building should be a GameObject with BuildingElement script.

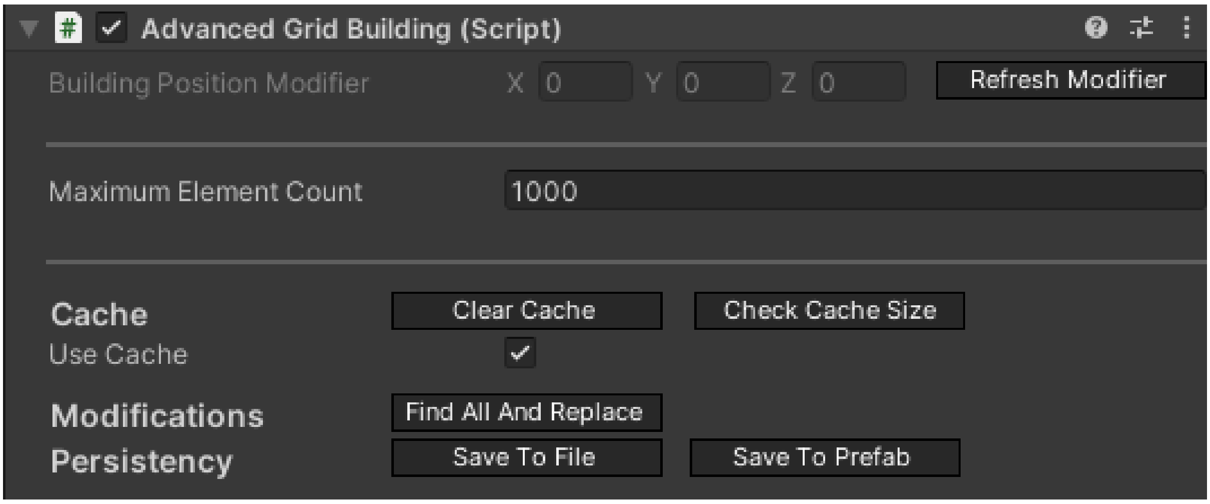
In case of Basic Gid Building and Free Building the algorithms use a global Building so the same Building will be used every time anywhere in the scene. For that the BuildingParent Object has dedicated Buildings for that case.

In the case of Advanced Grid Building or SnapPoint Based the algorithm will create a new Building if a new BuildingElement will be placed without snapping to another element. If an element is snapping to another one the new element will be placed under the snapping target's parent Building.

If every element is removed from a Building so it has left zero child BuildingElement the Building will be destroyed.

Property of the Building script

The Buildings have almost zero property but it has some just to ensure that if you make a prefab from a Building the required properties are saved into the prefab so these should be serializable.



At this moment it is not recommended to build a Building by hand. Build it in a test scene and use the Save To Prefab button on the Buildings' inspector view. This feature is available on every Building type. But if you create your Building ensure that the BuildingElements has a valid position. Also in the case of Advanced Grid Building ensure that you refresh the Modifier after you placed the first Element under the Building.

Maximum Element Count:

The BuildingElement's count can not be higher than the set number. If you snap an element to another one with a maxed out element count the building of the new element will be blocked. In case of DragBuilding the amount of elements above the maximum count. For example in case of 100 maximum element count and the Building has 90 elements then only 10 elements can be built so in case of drag building 10 elements will be allowed and every element above that 10 will be blocked.

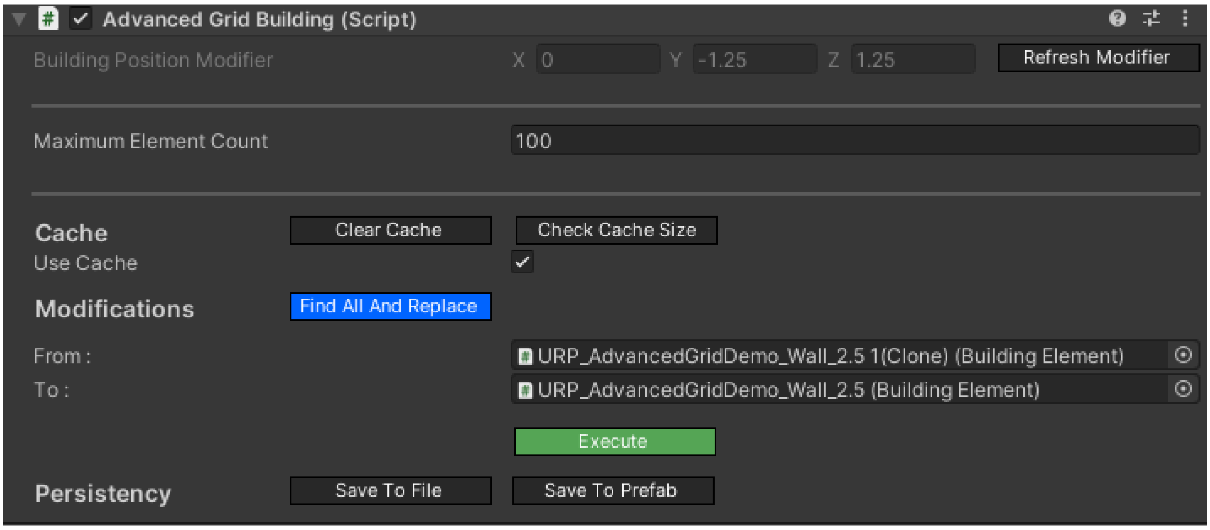
Cache

The Buildings are caching some data to increase the performance of the positioning algorithms. The cache increases the memory usage of the Buildings so the cache can be cleared, enabled or disabled any time.

Currently only the AdvancedGridBuilding and the BasicGridBuilding are actively using the cache.

Find All And Replace Elements

The Find and Replace button is an editor feature which will call the FindAllAndReplaceElements function on the building which is available during runtime if you need that functionality. The FindAllAndReplaceElements feature needs two BuildingElement. The “From” element will be searched in the children of the Building which will be replaced by the “To” element.



That it will replace every BuildingElements that match (the Guids are equal) with the provided “From” element.

The FindAllAndReplaceElements feature has some requirement :

- The provided BuildingElements must be not null.
- The elements can not be the same element so the guids have to be different.
- The elements’ main algorithm should be matching with the Building’s type
- The elements’ AdvancedGridType and AdvancedGridAxisType should be matching in case of AdvancedGridBuilding

If the requirements are not fulfilled the elements can’t be replaced.

Persistency

You can save the Building into a file with the Save To File. It will save a json file with the save data of the Building that is used during the save/load processes. You can use the BuildingLoader tool for loading a building from json data.

The FreeBuilding and the BasicGridBuildings are global Buildings which means that every time you place a BuildingElement with FreeBuilding or BasicGridBuilding anywhere in the world the element will be created under these objects. It will be explained later. The AdvancedGridBuilding is different because everytime when a new Building is started in the world a new AdvancedGridBuilding or a new SnapPointBasedBuilding is created to ensure that only those elements are under a Building that belongs to that Building.

IBuildingExternalInterface

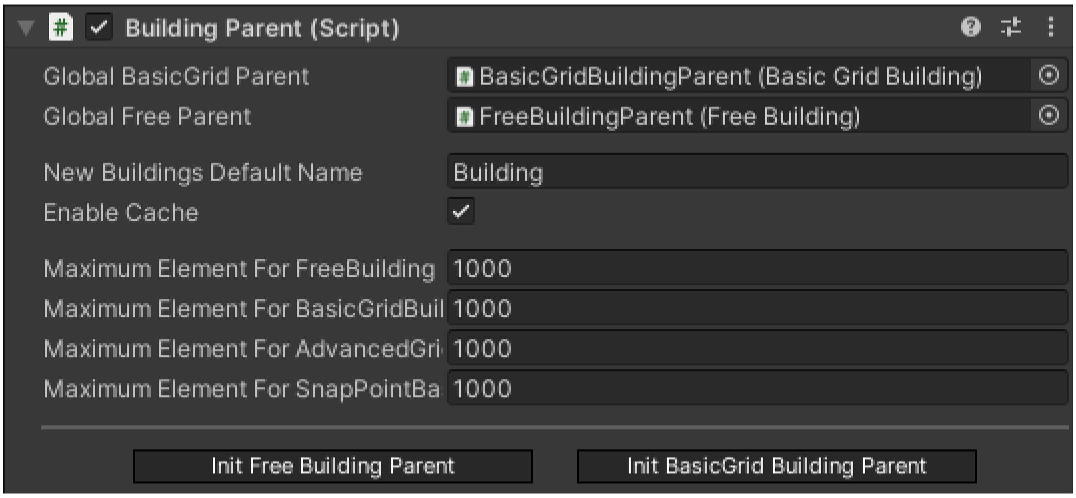
The IBuildingExternalInterface is a collection of functions that is available to the developers who's using this asset to deal with the Buildings. Every other public function is for the Advanced Building System. The goal of this interface is to help the developers and make it easier to know what options they have. Any other function call on the Buildings that is not included by this interface is not recommended.

Note that the Buildings are inherited from the SaveableMonobehaviour. Its function can be used on the Buildings to reach the persistence functionalities.

- **Getters/Setters**
    - LayerCollection
    - Elements
    - PositionSearchAlgorithmType
- **Add**
    - AddBuildingElement (2 overload)
        - With Position
        - With Position + Rotation
- **Remove**
    - RemoveBuildingElement
- **Search / Checks / Find**
    - FindBuildingElement
    - ContainsBuildingElement
    - GetPositionOfBuildingElement
    - FindAllBuildingElement (3 overload)
        - Based on BuildingElement's Guid
        - Based on BuildingElement's MetaData
        - Based on BuildingElement's AreaType
    - FindAllPreBuiltBuildingElement
    - FindAllFoundationBuildingElement
        - Inverse option = Find all not foundation
- **Element Modifications**
    - FindAllAndReplaceElements
    - MakePreBuilt
    - RemovePreBuilt
- **Material Modifications**
    - SetMaterialToDefault
    - SetMaterialBasedOnState
- **Cache**
    - ClearCache
    - EnableCache
    - DisableCache

BuildingParent

Every Building should be under a GameObject with a BuildingParent script. This script will manage and create the Buildings.



Global BasicGrid Parent is the global Basic Grid Building parent that is used for every new BuildingElement built with the BasicGrid Building algorithm.

Global Free Parent is the global Free Building parent that is used for every new BuildingElement built with the Free Building algorithm.

New Buildings Default Name will be the name of the newly created Buildings.

Enable Cache is a boolean variable and if it is true then the newly created Building's cache will be enabled. The cache will speed the algorithms up but increase the memory usage of the algorithms.

Maximum Element for … variables will be the maximum element count for the newly created Buildings for the specific type of Building.